

THE OSI MODEL

A Complete Guide to all Seven Layers

1. The OSI Model and why it matters

Imagine you want to send a parcel to someone in another country. You pack the item, seal the box, write the delivery address on it, hand it to the post office, and forget about the rest. From there, someone sorts it, a driver loads it into a van, a plane carries it across the ocean, another driver collects it, and a postman delivers it. You only packed the box. Every other step happened automatically, handled by a different part of a well-organized system where each person knows their job and only communicates with the person directly above or below them in the chain.

The OSI model works exactly the same way for data travelling across a network. It breaks the entire communication process into seven clearly defined layers. Each layer has one specific job. Each layer only talks to the layer directly above and below it. Together, they move data reliably from one device to another, anywhere in the world, regardless of the operating system, manufacturer, or type of hardware involved.

2. The Seven Layers at a Glance

The OSI model has seven numbered layers. Layer 1 sits at the bottom and handles the most basic physical transmission. Layer 7 sits at the top and is closest to the software applications people actually use.

THE 7 LAYERS OF THE OSI MODEL

7

APPLICATION

Works directly with users software. Services like HTTP, SMTP, FTP, DNS.

6

PRESENTATION

Formats, encrypts, and compresses data. Ensures cross-system compatibility and security.

5

SESSION

Establishes, maintains, and terminates communication sessions. Manages session security.

4

TRANSPORT

Ensures reliable and fast delivery using TCP & UDP. Handles segmentation, sequencing, and error control.

3

NETWORK

Routes packets across networks using IP addresses. Handles logical addressing.

2

DATA LINK

Ensures error-free data transmission. Provides node-to-node delivery using MAC addresses.

1

PHYSICAL

Transmits raw bits over physical media like fiber optics or radio waves. Defines standards for data signals.

3. Each Layer Explained in Full Details

Layer 7: Application Layer

This is the layer end users interact with directly. When you open a browser and type a URL, send an email, or transfer a file using FTP software, you are at the Application layer. This does not mean the application itself, rather the interface point where applications connect to the network and begin communicating with other devices.

The Application layer identifies communication partners, checks that network resources are available, and initiates the exchange of data. It provides the specific protocols that applications use to carry out their work.

Protocols that live at Layer 7:

- HTTP (port 80): The foundation of web communication. Used when loading any unencrypted webpage.
- HTTPS (port 443): The encrypted version of HTTP, secured with TLS underneath. Standard for all modern websites.
- FTP (ports 20 and 21): Transfers files between a client and a server. Port 21 for control, port 20 for data.
- SMTP (port 25): Sends outgoing email between mail servers.
- POP3 (port 110) and IMAP (port 143): Receive incoming email. POP3 downloads and deletes from server. IMAP keeps it on the server and syncs across devices.
- DNS (port 53): Translates human-readable domain names like microsoft.com into IP addresses.
- SSH (port 22): Encrypted remote access to servers and network devices. Replaced Telnet.
- Telnet (port 23): Unencrypted remote access. Still tested in certifications but avoided in practice.
- SNMP (ports 161 and 162): Monitors and manages network devices from a central location.

Real World Scenario:

When you type <https://www.google.com> into your browser and press Enter, your browser constructs an HTTPS request at Layer 7. That request travels down all seven layers on your device, crosses the network as electrical signals or radio waves, and climbs back up all seven layers on Google's server. The Application layer on their side reads your request and sends back a response. That response makes the same journey in reverse and becomes the page on your screen. Every time you load a webpage, this entire seven-layer journey happens usually in under 100 milliseconds.

Layer 6: Presentation Layer

The Presentation layer is the translator of the OSI model. Its job is to make sure that data sent by one system can actually be understood by another system, even when those two systems store and represent data in completely different internal formats.

Imagine you write a letter in English, and your recipient can only read French. Without someone to translate it before delivery, the letter is useless. The Presentation layer is that translator for data.

Three jobs at Layer 6:

1. **Translation:** Converts data between different character encoding formats, for example from EBCDIC (used by older IBM mainframes) to ASCII or Unicode (used by modern systems).
2. **Encryption and Decryption:** When you use HTTPS, the Presentation layer manages TLS encryption before data is passed down the stack, and decryption when data is received. This protects passwords, card numbers, and any sensitive data in transit.
3. **Compression:** Reduces the size of data before transmission to speed things up. Video platforms use MPEG, H.264, or H.265 encoding at this layer.

Common Layer 6 standards: SSL/TLS for encryption, JPEG and PNG for images, MPEG and H.264 for video, ASCII and Unicode for text.

Layer 5: Session Layer

The Session layer manages conversations between two applications. A session is an established connection that allows data to be exchanged over a period of time. The Session layer sets it up, keeps it alive, and closes it cleanly when both sides are finished.

Real world Scenario: During a phone call, one person dials, the other picks up (session establishment). They talk for as long as needed (session maintenance). One person says goodbye and hangs up (session termination).

The Session layer also handles checkpointing. In a long data transfer, checkpoints are inserted at intervals. If the connection drops at 80% completion, the transfer can resume from the last checkpoint instead of restarting from zero. This is how large file downloads can continue after a brief internet interruption.

Session layer technologies: NetBIOS for local network sessions, RPC (Remote Procedure Call) for cross-device service requests, SIP (Session Initiation Protocol) for VoIP voice and video calls.

Layer 4: Transport Layer

The Transport layer is where end-to-end delivery is managed. It takes data from the upper layers and breaks it into smaller pieces called segments. It makes sure those segments reach the correct application on the destination device, reassembles them in the correct order on arrival, and retransmits any that were lost.

Port numbers operate at this layer. They are the mechanism that tells a device which application should receive a given piece of data. For example, traffic arriving on port 443 goes to the web server process. Traffic on port 22 goes to the SSH daemon. This is how a single device can run many network services simultaneously without confusion.

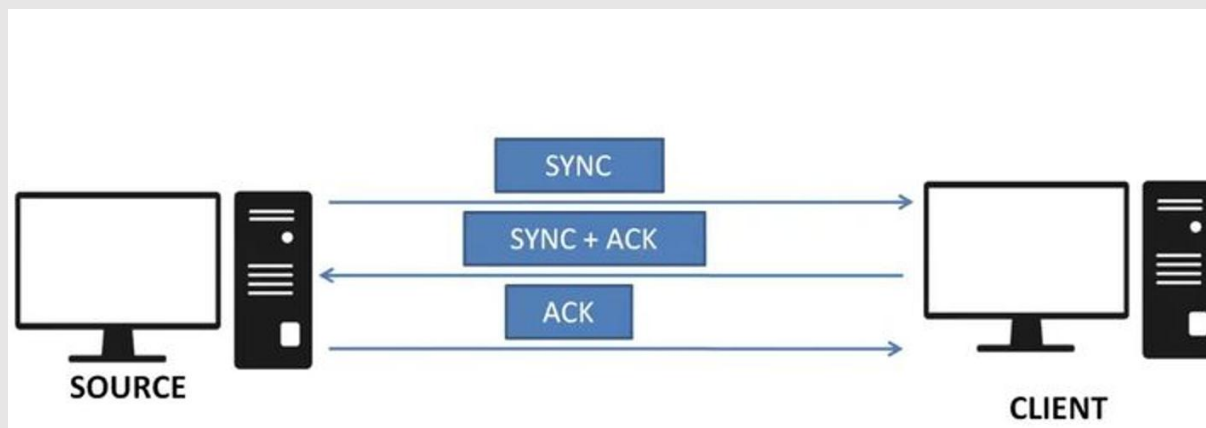
Feature	TCP	UDP
Connection type	Connection-oriented. Handshakes before sending.	Connectionless. Sends immediately.
Reliability	Guaranteed. Lost packets are retransmitted.	No guarantee. Packets can be dropped.
Speed	Slower due to acknowledgement overhead.	Faster. No overhead.
Packet ordering	Arrives in correct order.	May arrive out of order.
Error checking	Full error detection and correction.	Checksum only.
Header size	20 bytes minimum.	8 bytes.
Typical use case	Web browsing, email, file transfer, SSH.	Streaming, DNS, VoIP, online gaming.

The TCP Three-way Handshake

Before TCP sends a single byte of actual data, it establishes a connection through a three-step process. Both sides agree they are ready, confirm they can reach each other, and then data transfer begins.

Step	Who Acts	Message Sent	What It Means in Plain Language
1	Sender	SYN	I want to start a conversation. Are you available? (Synchronize)
2	Receiver	SYN-ACK	Yes, I am here and ready. I acknowledge your request. (Synchronize-Acknowledge)
3	Sender	ACK	Understood. Connection confirmed. Let's begin. (Acknowledge)

This process happens for every TCP connection, including every webpage you load. When the data transfer is complete, a four-step teardown process closes the connection cleanly on both sides.



TCP three-way handshake

Layer 3: Network Layer

The Network layer moves data from one network to a completely different network. While Layer 2 handles delivery within a single local network, Layer 3 handles delivery across many interconnected networks, including the entire internet.

The two core responsibilities at Layer 3 are logical addressing and routing. Logical addressing means assigning IP addresses to devices. These addresses can be changed and organized into hierarchical groups called subnets. Routing means determining the best path for data to travel from its source network to its destination, potentially crossing dozens of different networks along the way.

The device that operates at Layer 3 is the router. Every time your data crosses from your home network to your ISP, or from your ISP toward a server somewhere in the world, it passes through routers making Layer 3 decisions.

Key Layer 3 protocols:

- **IP:** The Internet Protocol. Every device needs an IP address. IPv4 uses 32-bit addresses. IPv6 uses 128-bit addresses.
- **ICMP:** Internet Control Message Protocol. Used by ping and traceroute for diagnostics and error reporting.
- **OSPF:** Open Shortest Path First. A routing protocol is used inside enterprise networks to find efficient routes.

- BGP: Border Gateway Protocol. The routing protocol that runs the internet. Controls how traffic moves between large networks operated by ISPs and organizations.

Layer 2: Data Link Layer

The Data Link layer handles communication between devices that share the same local network. While Layer 3 uses IP addresses to route between networks, Layer 2 uses MAC addresses to deliver frames from one device to another within a single network segment.

A MAC address is a unique hardware identifier built into every network interface card during manufacture. It is 48 bits long, written in hexadecimal like 00:1A:2B:3C:4D:5E. Unlike IP addresses, MAC addresses are not routable across the internet. They are only used for local delivery.

The Data Link layer splits into two sublayers:

- LLC (Logical Link Control): Handles flow control and error checking between adjacent devices. Defined by IEEE 802.2.
- MAC (Media Access Control): Controls how devices access the shared medium and handles hardware addressing. Ethernet (IEEE 802.3) and Wi-Fi (IEEE 802.11) operate here.

The primary device at Layer 2 is the switch. A switch learns which MAC address is connected to which port and sends frames only to the correct port, making it far more efficient than a hub, which broadcasts to every port.

Practical Application:

When your device wants to send data to another device on the same network, it knows the destination IP address but needs the MAC address to build a Layer 2 frame. ARP broadcasts a question to the entire local network: Who has IP 192.168.1.1? The device with that IP responds with its MAC address. Your device stores this in its ARP cache so it does not have to ask again for a while. You can view your ARP cache on Windows by running `arp -a` in the command prompt.

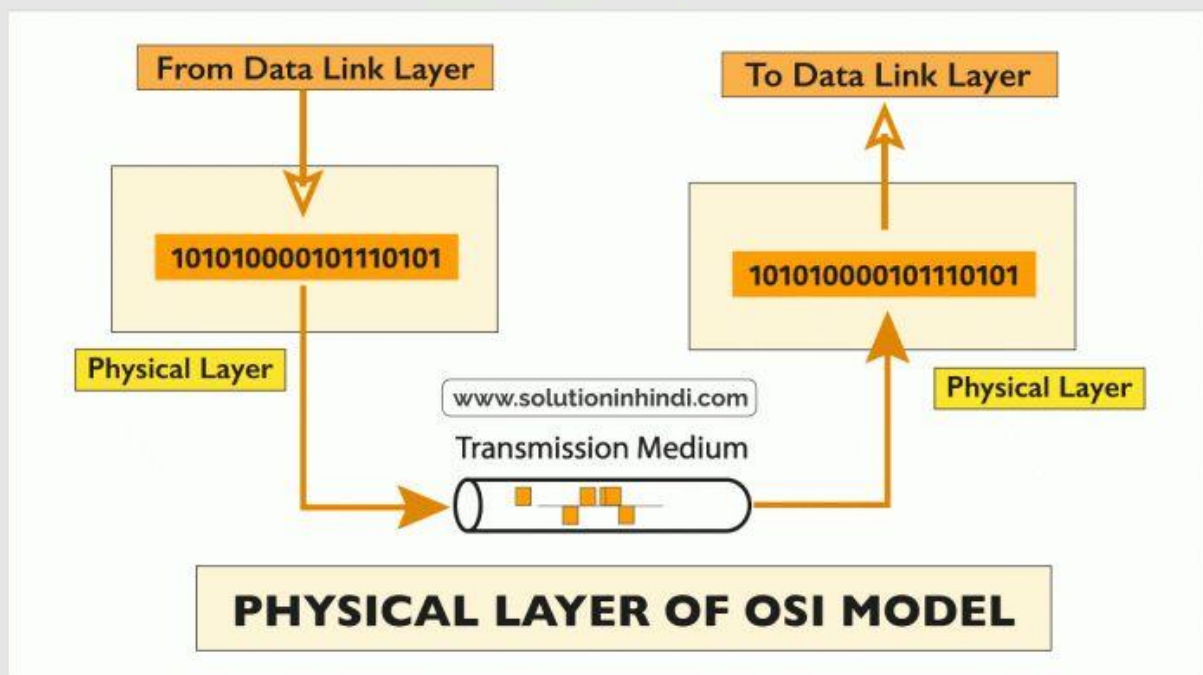
Physical Layer: Layer 1

The Physical layer is the foundation of the entire model. By the time data reaches Layer 1, it is no longer a packet or a frame. It is a raw stream of binary 1s and 0s, and the Physical layer's job is to convert those bits into a signal suitable for the physical medium being used.

For a copper cable, bits become electrical voltage signals. For fibre optic cable, bits become pulses of light. For a wireless connection, bits become radio waves. The Physical layer defines the rules for how signals are encoded, what speeds are supported, and what the physical connectors look like.

Physical layer components:

- Cables: Cat 5e, Cat 6, Cat 6a, and Cat 7 for copper. Single-mode and multi-mode fibre for optical.
- Connectors: RJ-45 for Ethernet. LC, SC, and ST for fibre optic.
- Wireless: 2.4 GHz and 5 GHz radio bands used by Wi-Fi standards 802.11n, 802.11ac, and 802.11ax.
- Layer 1 devices: Hubs, repeaters, and cables. These have no intelligence; they simply move signals.



4. Encapsulation and Decapsulation

Encapsulation is the process of packaging data as it moves down through the seven layers on the sending device. Decapsulation is the reverse: stripping those packages off as data moves up through the layers on the receiving device.

Think of packing a gift for international shipping. You place the gift in a box (Layer 7). You add foam padding for protection (Layer 6 encrypts it). You seal it and write tracking information (Layer 5 session info). You address the outer box with delivery details (Layers 3 and 2 add IP and MAC headers). The courier physically transports it (Layer 1). When the recipient opens it, they remove each layer of packaging in reverse order until they reach the original gift. That unwrapping is decapsulation.

Layer(s)	Name	PDU	What Gets Added to the Data
7, 6, 5	Application / Presentation / Session	Data	Original message created, formatted, encrypted, session established
4	Transport	Segment	TCP or UDP header: source port, destination port, sequence number
3	Network	Packet	IP header: source IP, destination IP, TTL value
2	Data Link	Frame	MAC header and FCS trailer: source MAC, destination MAC, error check
1	Physical	Bits	Converted to electrical signals, light pulses, or radio waves

On the receiving device, each layer reads only its own header, strips it off, and passes the remaining data upward. By the time data reaches Layer 7 at the destination, it is identical to what was created at Layer 7 at the source.

5. The OSI vs the TCP/IP Model

You will hear about the TCP/IP model just as often as OSI. They describe the same communication process but with different levels of detail.

OSI has seven layers and is a conceptual reference model used for teaching and troubleshooting. No real protocol stack maps perfectly to all seven layers in practice. The TCP/IP model has four layers and describes how the internet actually works. All real internet communication uses TCP/IP.

OSI Layer	TCP/IP Equivalent
7 — Application	Application (absorbs Layers 5, 6, and 7)
6 — Presentation	(merged into Application layer)
5 — Session	(merged into Application layer)
4 — Transport	Transport
3 — Network	Internet
2 — Data Link	Network Access (absorbs Layers 1 and 2)
1 — Physical	(merged into Network Access layer)

OSI gives you the granular detail needed to understand and troubleshoot networking deeply. TCP/IP reflects the real world. You need to be fluent in both.

6. Ports You Must Know

Port numbers operate at Layer 4. Think of an IP address as a building's street address and a port number as the apartment number inside. The IP address gets data to the right device. The port number gets it to the right application on that device.

Ports range from 0 to 65535. Ports 0 to 1023 are well-known ports reserved for standard services. Ports 1024 to 49151 are registered ports used by specific applications. Ports 49152 to 65535 are ephemeral ports assigned temporarily by the operating system for outgoing connections.

Port	Protocol	What It Is Used For
20, 21	FTP	File transfer. Port 21 for commands, port 20 for data.
22	SSH	Encrypted remote access to servers and devices. Replaced Telnet.
23	Telnet	Unencrypted remote access. Largely replaced by SSH.
25	SMTP	Sends outgoing email between mail servers.
53	DNS	Resolves domain names to IP addresses. Uses UDP mostly, TCP for large responses.
67/68	DHCP	Server listens on 67. Client sends from 68. Assigns IP addresses automatically.
80	HTTP	Unencrypted web traffic.
110	POP3	Downloads email from server to local device.
143	IMAP	Reads email while keeping it on the server. Supports multiple devices.
443	HTTPS	Encrypted web traffic using TLS.
3389	RDP	Windows Remote Desktop. Graphical remote access.

7. Troubleshooting with the OSI Model

The single most practical benefit of knowing the OSI model is the ability to troubleshoot network problems systematically instead of guessing. The standard approach for a complete network failure is bottom-up: start at Layer 1 and work upward until you find the problem.

Layer	Name	What to Check and Ask
1	Physical	Is the cable connected? Is the link light on? Try a different cable or port on the switch.
2	Data Link	Can the switch see the device? Is the MAC in the switch table? Is there a duplex mismatch?
3	Network	Does the device have a valid IP? Can it ping the default gateway? Are routes correct?
4	Transport	Is the port open? Is a firewall blocking it? Use netstat to see which services are listening.
5	Session	Are sessions forming? Are they timing out? Are authentication failures preventing sessions?
6	Presentation	Are SSL/TLS certificates valid and not expired? Are there encoding or format mismatches?
7	Application	Is the service running? Is DNS resolving correctly ? Check application-level error logs.

Real World Scenario:

A junior engineer once spent three hours troubleshooting a PC that had no network access. He restarted services, reconfigured IP addresses, and reinstalled the network driver twice. A senior engineer walked over, looked at the back of the machine, and noticed the Ethernet cable was seated in a switch port that had been administratively shut down. Layer 1. Fixed in two seconds. The lesson: always start at the bottom before changing anything at the top.

8. OSI and Azure: Where Cloud Services Fit

For anyone working in Azure administration or cloud infrastructure, knowing where Azure services sit in the OSI model is directly relevant to exam questions and daily job tasks.

Azure Service	OSI Layer	What It Does
Azure Virtual Network (VNet)	Layer 3	IP addressing, subnetting, and routing for Azure resources.
Network Security Groups (NSGs)	Layers 3 and 4	Filter traffic using IP addresses and port numbers.
Azure Load Balancer	Layer 4	Distributes TCP and UDP traffic across backend virtual machines.
Azure Application Gateway	Layer 7	HTTP and HTTPS load balancer. Routes traffic by URL path and host headers.
Azure Firewall	Layers 3 to 7	Inspects and filters traffic from network layer up to application layer.
Azure DNS	Layer 7	Name resolution for Azure resources and custom domains.
Azure VPN Gateway	Layer 3	Extends on-premises networks into Azure over encrypted tunnels.
Azure ExpressRoute	Layer 2	Private dedicated fibre connection from on-premises to Azure.